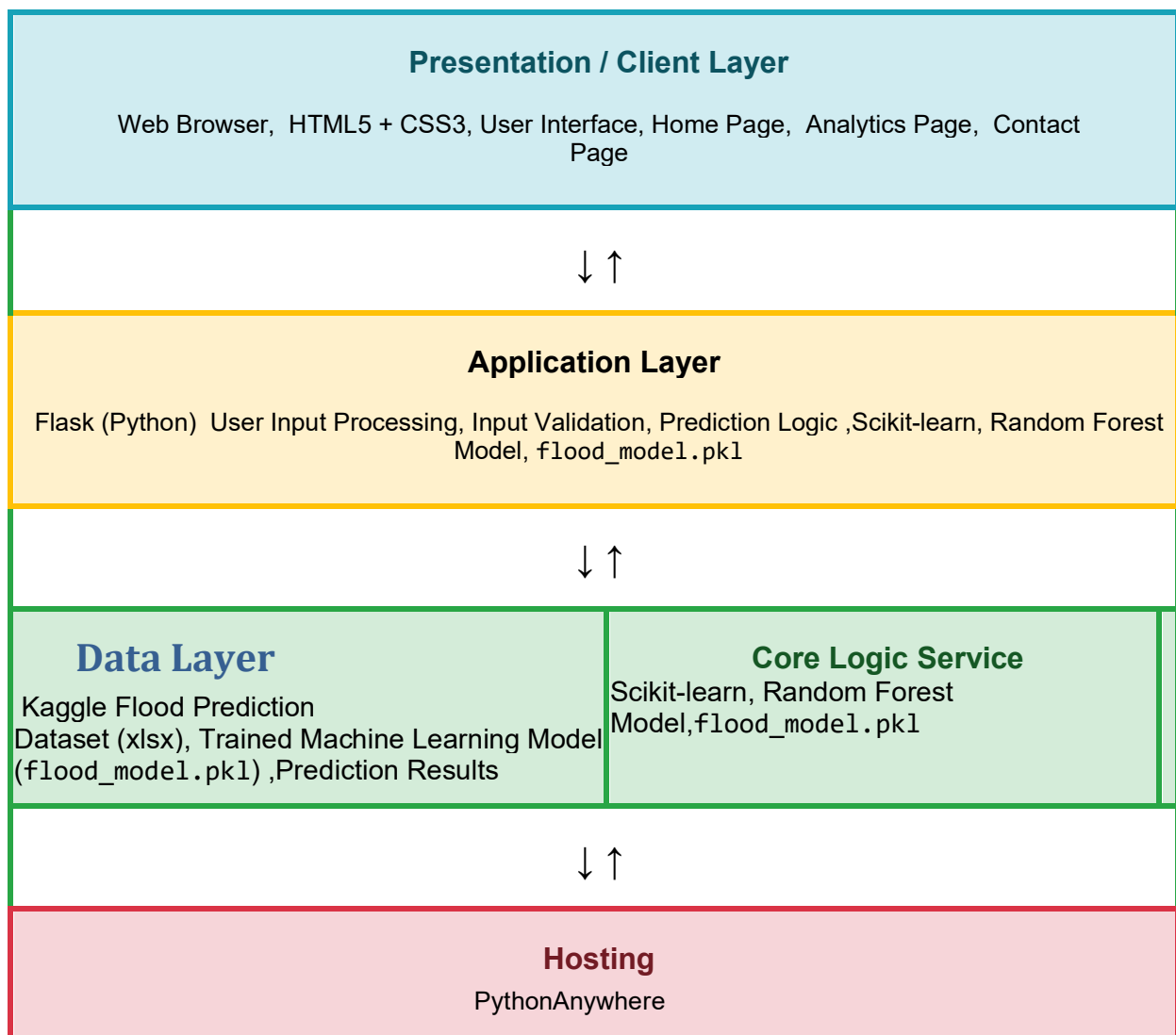


## Solution Architecture

|                      |               |
|----------------------|---------------|
| <b>Date</b>          | 30 June 2026  |
| <b>Team ID</b>       | XXXXXX        |
| <b>Project Name</b>  | Rising Waters |
| <b>Maximum Marks</b> | 5 Marks       |

### Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



## Component Description Table

| Component Name         | Description / Role in Architecture                                                                                                               | Technologies Used                                     |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Presentation Layer     | Provides a user-friendly web interface for entering weather parameters, viewing flood predictions, analytics, and emergency contact information. | HTML5, CSS3                                           |
| Application Layer      | Processes user requests, validates inputs, loads the trained machine learning model, and generates flood predictions.                            | Flask (Python), Scikit-learn                          |
| Machine Learning Layer | Performs flood risk prediction using the trained Random Forest machine learning model.                                                           | Random Forest, Scikit-learn, Pickle (flood_model.pkl) |
| Data Layer             | Stores the dataset used for model training and the trained machine learning model for prediction.                                                | Kaggle Flood Prediction Dataset (xlsx), Pickle (.pkl) |